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CS61C

Great Ideas in Computer Architecture

(a.k.a. Machine Structures)



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Introduction to Synchronous Digital Systems (SDS) State

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Accumulator

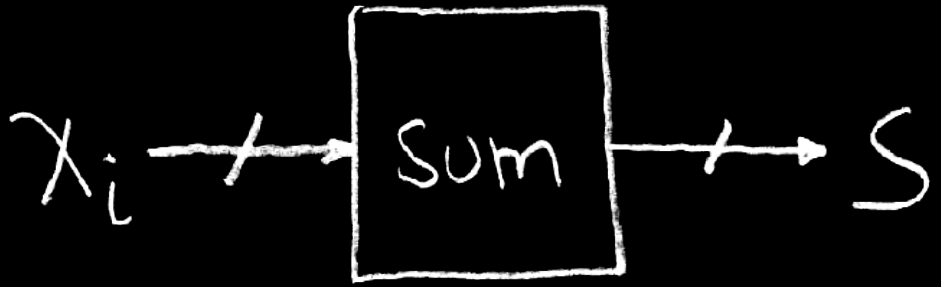
Uses for State Elements

- **As a place to store values for some indeterminate amount of time:**
  - Register files (like `x0-x31` on the RISC-V)
  - Memory (caches, and main memory)
- **Help control the flow of information between combinational logic blocks.**
  - State elements are used to hold up the movement of information at the inputs to combinational logic blocks and allow for orderly passage.

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Accumulator Example

- **Why do we need to control the flow of information?**



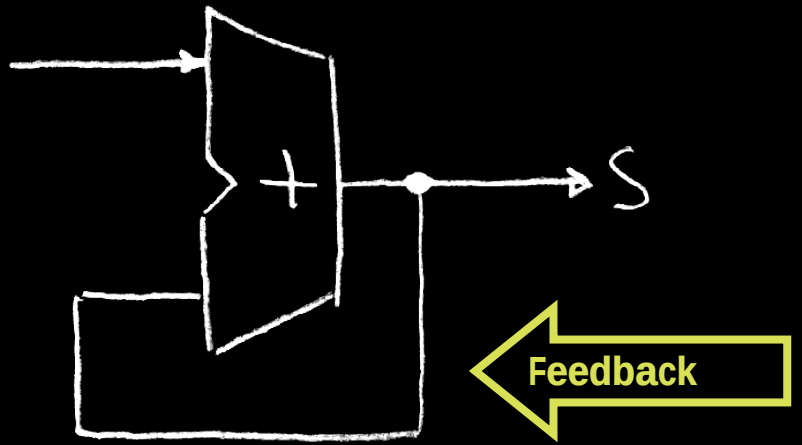
- **Want:**

```
S=0;
for (i=0;i<n;i++)
   S = S + Xi
```

- **Assume**
  - Each **X** value is applied in succession, one per cycle.
  - After **n** cycles the sum is present on **S**.

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First try…Does this work?

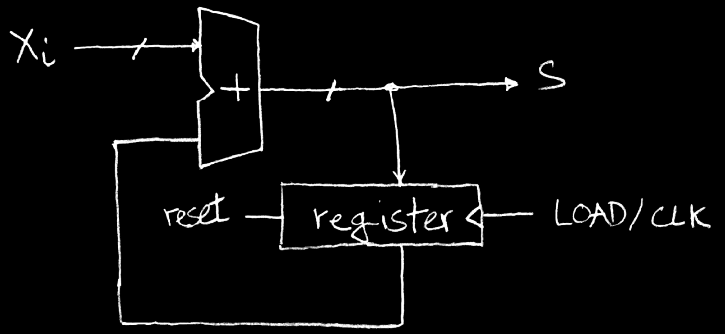


- **Nope!**
  - Reason #1… What is there to control the next iteration of the '`for`' loop?
  - Reason #2… How do we say: '`S=0`'?

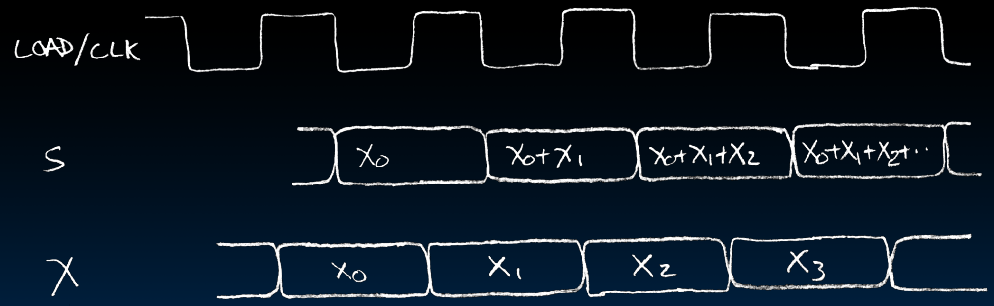
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Second try…How about this?

- Register is used to hold up the transfer of data to adder.



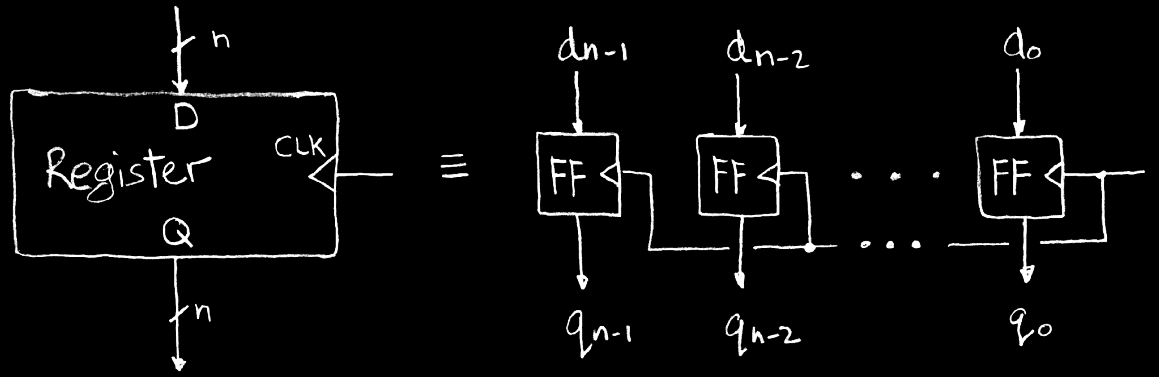
Rough timing…



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Register Details
Flip-flops

Register Details…What's inside?

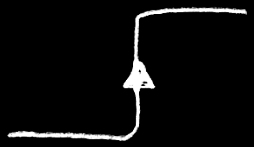


- n instances of a "Flip-Flop"
- Flip-flop name because the output flips and flops between and 0,1
- D is "data", Q is "output"
- Also called "D-type Flip-Flop"
  - There used to be other types of flip-flops

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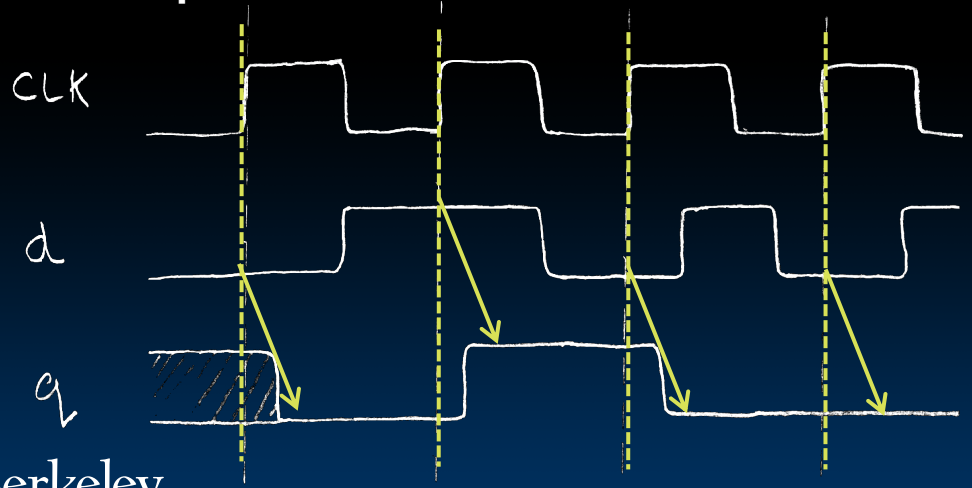
What's the timing of a Flip-flop? (1/2)

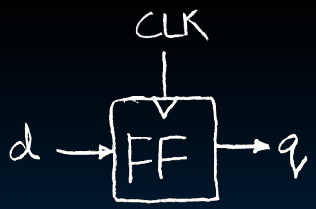
- **Edge-triggered d-type flip-flop**
  - This one is "rising edge-triggered"
  - Also called "positive edge"



There also exist "falling edge" FFs

- **"On the rising edge of the clock, the input d is sampled and transferred to the output. At all other times, the input d is ignored."**
- **Example waveforms:**

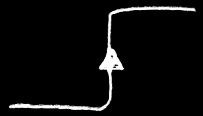




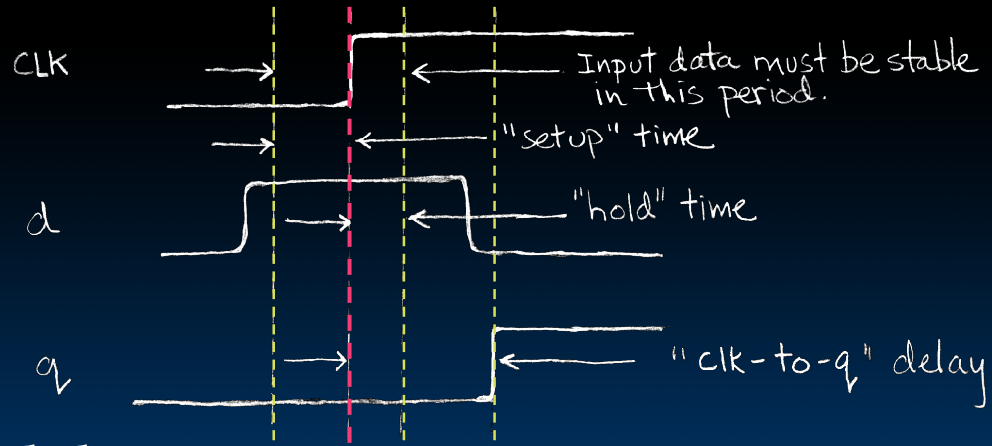
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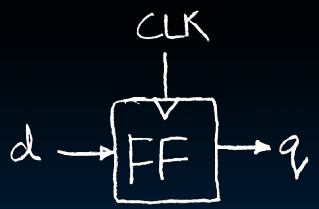
What's the timing of a Flip-flop? (2/2)

- **Edge-triggered d-type flip-flop**
  - This one is "rising edge-triggered"



- **"On the rising edge of the clock, the input d is sampled and transferred to the output. At all other times, the input d is ignored."**
- **Example waveforms (more detail):**



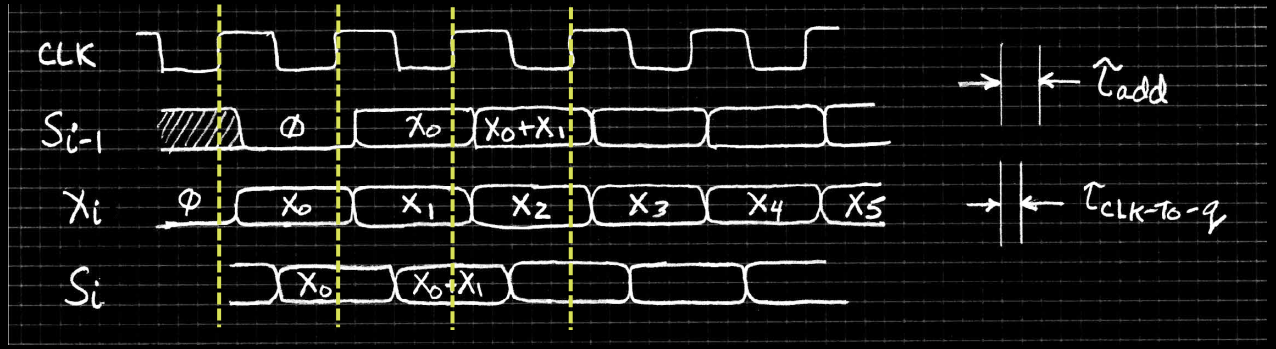


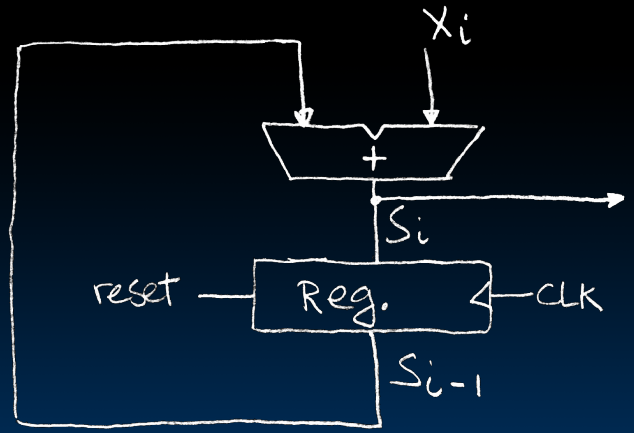


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Accumulator Revisited

Accumulator Revisited (proper timing 1/2)

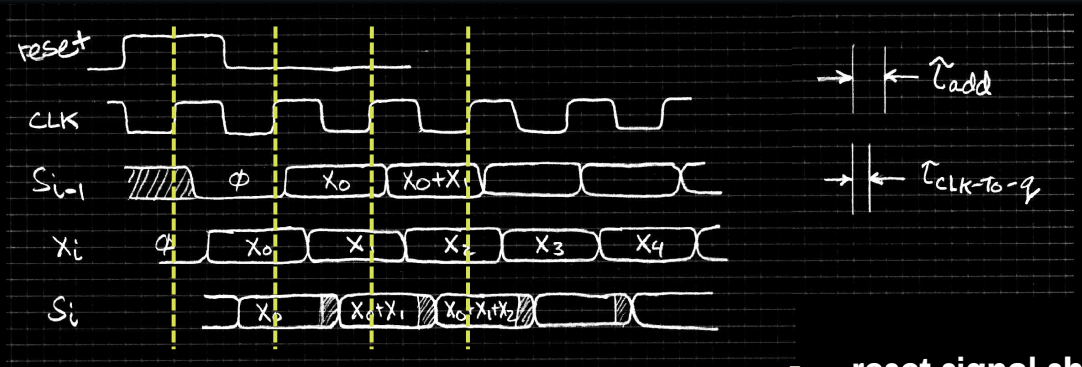


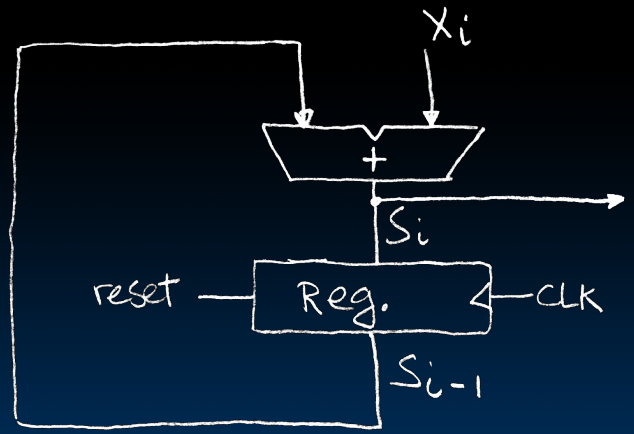


- Reset input to register is used to force it to all zeros (takes priority over D input).
- S_{i-1} holds the result of the ith-1 iteration.
- Analyze circuit timing starting at the output of the register.

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Accumulator Revisited (proper timing 2/2)





- reset signal shown.
- Also, in practice X might not arrive to the adder at the same time as S_{i-1}
- S_i temporarily is wrong, but register always captures correct value.
- In good circuits, instability never happens around rising edge of clk.

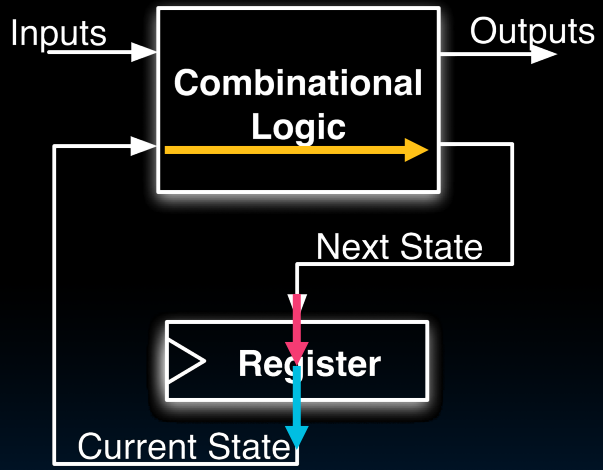


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Pipelining for Performance

Maximum Clock Frequency

- What is the maximum clock frequency of this circuit? (Hint: Frequency = 1 / Period)

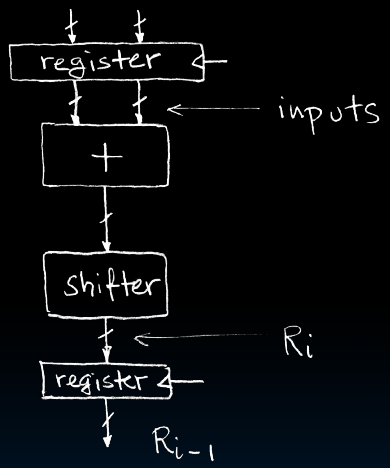


Max Delay = CLK-to-Q Delay + CL Delay + Setup Time

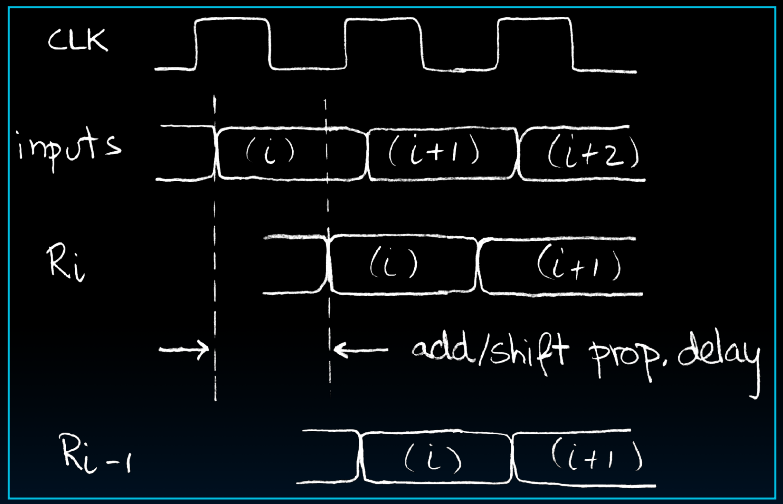
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Pipelining to improve performance (1/2)

- **Extra Registers** are often added to help speed up the clock rate.



Timing…

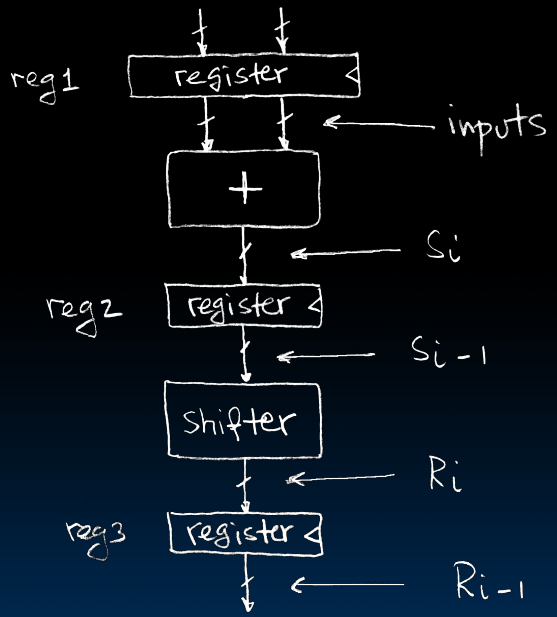


- **Note: Delay of 1 clock cycle from input to output.**
- **Clock period limited by propagation delay of adder/shifter.**

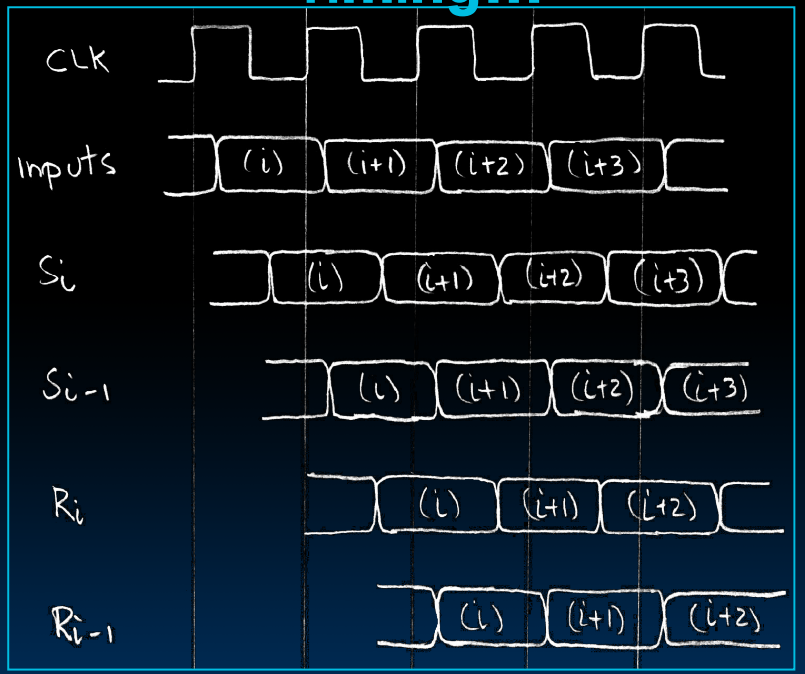
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Pipelining to improve performance (2/2)

- Insertion of register allows higher clock frequency.
- More outputs per second.



Timing…



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Recap of Timing Terms

- **Clock (CLK)** - steady square wave that synchronizes system
- **Setup Time** - when the input must be stable before the rising edge of the CLK
- **Hold Time** - when the input must be stable after the rising edge of the CLK
- **"CLK-to-Q" Delay** - how long it takes the output to change, measured from the rising edge of the CLK
- **Flip-flop** - one bit of state that samples every rising edge of the CLK (positive edge-triggered)
- **Register** - several bits of state that samples on rising edge of CLK or on LOAD (positive edge-triggered)

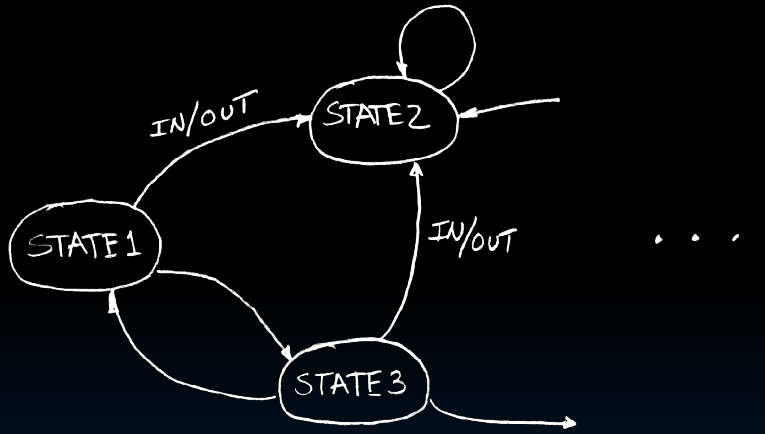


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Finite State Machines

Finite State Machines (FSM) Introduction

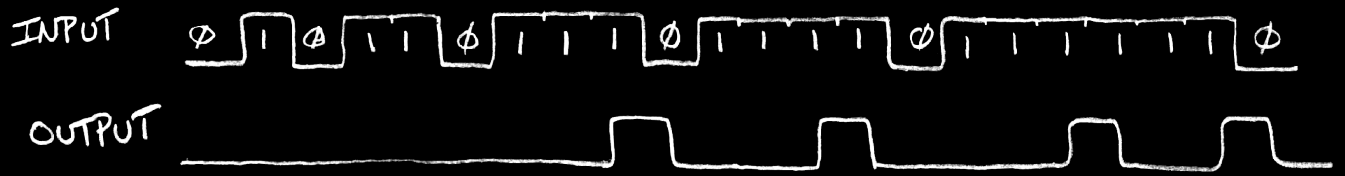
- You have seen FSMs in other classes
  - Same basic idea
- The function can be represented with a "state transition diagram"
- With combinational logic and registers, any FSM can be implemented in hardware.



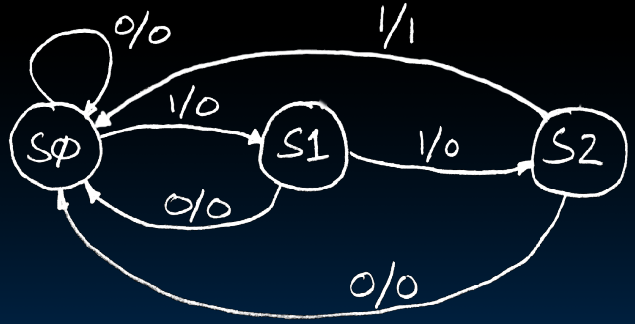
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Finite State Machine Example: 3 ones…

- **FSM to detect the occurrence of 3 consecutive 1's in the input.**



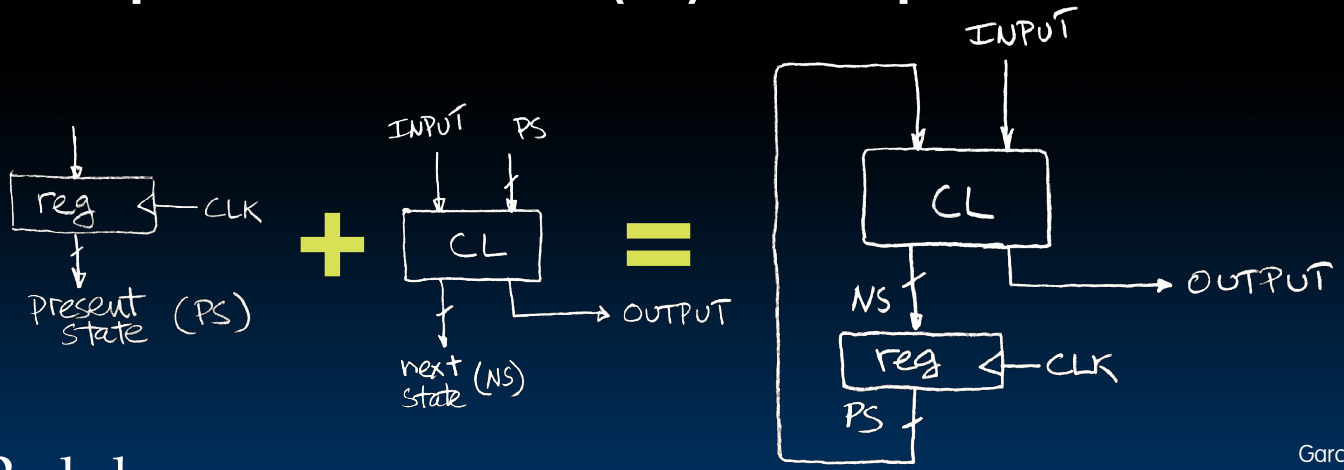
- **Draw the FSM…**
  - Assume state transitions are controlled by the clock: on each clock cycle the machine checks the inputs and moves to a new state and produces a new output…



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Hardware Implementation of FSM

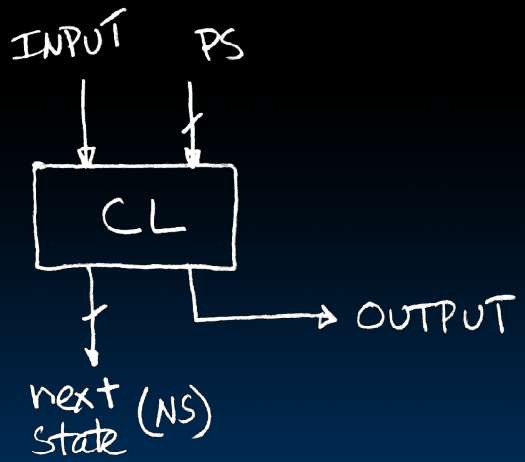
- … Therefore a register is needed to hold the representation of which state the machine is in.
  - Use a unique bit pattern for each state.
- Combinational logic circuit is used to implement a function mapping the input and present state (PS) input to the next state (NS) and output.



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Hardware for FSM: Combinational Logic

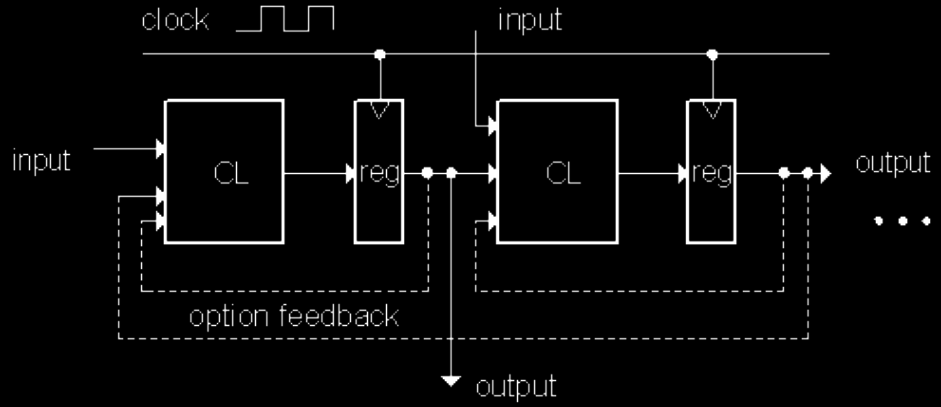
- Next lecture we will discuss the detailed implementation, but for now can look at its functional specification, truth table form.



PS	Input	NS	Output
00	0	00	0
00	1	01	0
01	0	00	0
01	1	10	0
10	0	00	0
10	1	00	1

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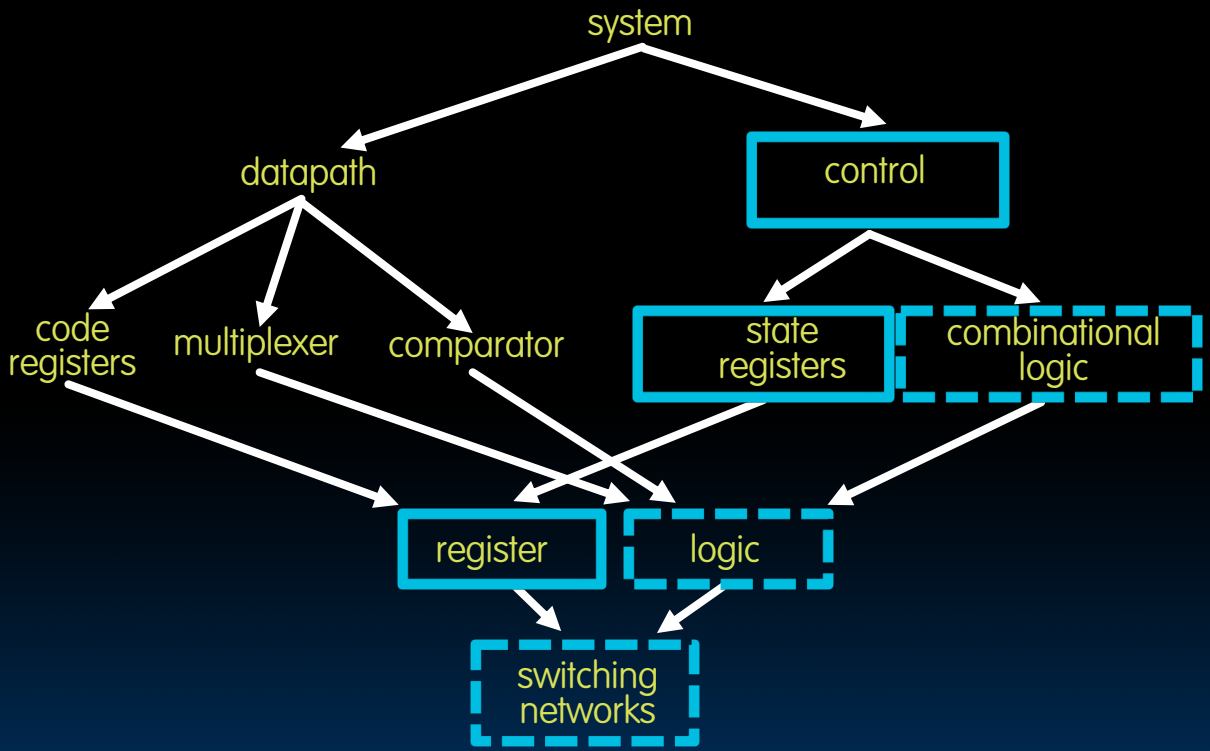
General Model for Synchronous Systems



- Collection of CL blocks separated by registers.
- Registers may be back-to-back and CL blocks may be back-to-back.
- Feedback is optional.
- Clock signal(s) connects only to clock input of registers.

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Design Hierarchy



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L15 HW feedback akin to SW recursion | The minimum period of a usable synchronous circuit is at least the setup + CLK-to-Q delay | You can build a FSM to signal when an equal number of 0s and 1s has appeared in the input.

- FFF
- FFT
- FTF
- FTT
- TFF
- TFT
- TTF
- TTT

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"And In conclusion…"

- **State elements are used to:**
  - Build memories
  - Control the flow of information between other state elements and combinational logic
- **D-flip-flops used to build registers**
- **Clocks tell us when D-flip-flops change**
  - setup and hold times are important
- **We pipeline long-delay CL for faster clock**
- **Finite state machines extremely useful**
  - You'll see them again 151A, 152, 164, 172, …



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